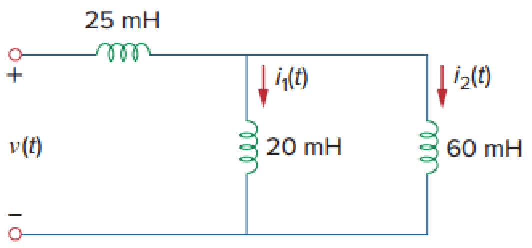
Chapter 6, Problem 62P

Problem

Consider the circuit in Fig. 6.84. Given that $v(t)$ = 12e−3t mV for t > 0 and i1(0) = −30 mA, find: (a) i2(0), (b) i1(t) and i2(t).

Fig. 6.84:



Step-by-step solution

Step 1 of 8

Refer to Figure 6.84 in the textbook.

Observe that the 20 mH and 60 mH inductors are connected in parallel combination.

Determine the equivalent inductance of 20 mH and 60 mH.

$$\begin{aligned} L_{eq1} &= \frac{(20\text{ mH})(60\text{ mH})}{20\text{ mH}+60\text{ mH}} \\ &= \frac{(20\text{ mH})(60\text{ mH})}{80\text{ mH}} \\ &= 15\text{ mH} \end{aligned}$$

Now, the resultant inductance 15 mH and 25 mH are connected in series combination.

Determine the equivalent inductance of 25 mH and 15 mH.

$$\begin{aligned} L_{eq} &= L_1 + L_2 \\ &= 25\text{ mH} + 15\text{ mH} \\ &= 40\text{ mH} \end{aligned}$$

Step 2 of 8

Calculate the total current through the circuit.

$$i(t) = \frac{1}{L_{eq}}\int_0^t v(t)\,dt + i(0)$$

Consider that the supply voltage, $v(t) = 12e^{-3t}\text{ mV}$.

Substitute $12e^{-3t}\text{ mV}$ for $v(t)$, 40 mH for L_{eq}.

$$\begin{aligned} i(t) &= \frac{1}{40\times10^{-3}}\int_0^t \left(12\times10^{-3}e^{-3t}\right)dt + i(0) \\ &= \frac{12\times10^{-3}}{40\times10^{-3}}\int_0^t e^{-3t}\,dt + i(0) \\ &= \frac{12\times10^{-3}}{40\times10^{-3}}\left(\frac{e^{-3t}}{-3}\right)_0^t + i(0) \\ &= \frac{-0.3}{3}\left(e^{-3t} - e^{-0}\right) + i(0) \end{aligned}$$

$$i(t) = -0.1\left(e^{-3t} - 1\right) + i(0)$$

Step 3 of 8

Use current division method to determine the current through 20 mH inductor.

$$\begin{aligned} i_1(t) &= \left(\frac{60\ \text{mH}}{60\ \text{mH} + 20\ \text{mH}}\right) i(t) \\ &= \left(\frac{60}{80}\right) i(t) \\ &= \frac{3}{4} i(t) \end{aligned}$$

Similarly, use current division method to determine the current through 60 mH inductor.

$$\begin{aligned} i_2(t) &= \left(\frac{20\ \text{mH}}{60\ \text{mH} + 20\ \text{mH}}\right) i(t) \\ &= \left(\frac{20}{80}\right) i(t) \\ &= \frac{1}{4} i(t) \end{aligned}$$

Step 4 of 8

Determine the current $i(0)$.

Take the equation of $i_1(t)$ and solve for $i(t)$.

$$i_1(t) = \frac{3}{4} i(t)$$
$$i(t) = \frac{4}{3} i_1(t)$$

Substitute 0 for t.

$$i(0) = \frac{4}{3} i_1(0)$$

Consider that the initial current across 20 mH inductor, $i_1(0) = -30\ \text{mA}$.

Substitute $-30\ \text{mA}$ for $i_1(0)$.

$$\begin{aligned} i(0) &= \frac{4}{3}(-30\ \text{mA}) \\ &= -40\ \text{mA} \end{aligned}$$

Step 5 of 8

(a)

Calculate the current $i_2(0)$.

Take the equation of $i_1(t)$ and substitute 0 for t.

$$i_2(t) = \frac{1}{4} i(t)$$
$$i_2(0) = \frac{1}{4} i(0)$$

Substitute $-40\ \text{mA}$ for $i(0)$.

$$\begin{aligned} i_2(0) &= \frac{1}{4}(-40\ \text{mA}) \\ &= -10\ \text{mA} \end{aligned}$$

Thus, the value of $i_2(0)$ is $\boxed{-10\ \text{mA}}$.

Step 6 of 8

(b)

Calculate the current $i(t)$.

$$i(t) = -0.1\left(e^{-3t} - 1\right) + i(0)$$

Substitute $-40\ \text{mA}$ for $i(0)$.

$$\begin{aligned} i(t) &= -0.1\left(e^{-3t} - 1\right) - 40\ \text{mA} \\ &= -0.1e^{-3t} + 0.1 - 0.04 \\ &= -0.1e^{-3t} + 0.06 \\ &= \left(-100e^{-3t} + 60\right)\ \text{mA} \end{aligned}$$

Step 7 of 8

Determine the current $i_1(t)$.

$$i_1(t) = \frac{3}{4} i(t)$$

Substitute the expression of $i(t)$.

$$\begin{aligned} i_1(t) &= \frac{3}{4}\left(-100e^{-3t} + 60\right)\ \text{mA} \\ &= \left(-75e^{-3t} + 45\right)\ \text{mA} \end{aligned}$$

Thus, the current through 20 mH inductor is $\boxed{\left(-75e^{-3t} + 45\right)\ \text{mA}}$.

Step 8 of 8

Determine the current $i_2(t)$.

$$i_2(t) = \frac{1}{4} i(t)$$

Substitute the expression of $i(t)$.

$$\begin{aligned} i_2(t) &= \frac{1}{4}\left(-100e^{-3t} + 60\right)\ \text{mA} \\ &= \left(-25e^{-3t} + 15\right)\ \text{mA} \end{aligned}$$

Thus, the current through 20 mH inductor is $\boxed{\left(-25e^{-3t} + 15\right)\ \text{mA}}$.